Topic 0: Syllabus Overview

ECE 9433: Agile System-on-Chip Design

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Prof. Austin Rovinski

New York University

What is this Course About?

- System-on-Chip
- Design
- Agile

Course Goals

- Imagine you are creating a small hardware startup company
  - Or academic/government/industry lab, hardware division of large company
- Understand how to create a specialized hardware solution from conception all the way through engineering prototype
  - Prototypes are what investors / stakeholders / academic reviewers want to see for funding
- Business / marketing aspects: see entrepreneurship courses

The SoC Design Principle

- The design process is a tradeoff of one or more of the following:
  - Engineer time
  - Computation time
  - Modeling accuracy/verification
  - Optimization

Prerequisites

- ECE-GY 6473 Introduction to VLSI Design
  - Fundamentals of MOSFETs, interconnects, primitive devices
- ECE-GY 6443 VLSI System and Architecture
  - Combinational & sequential logic, Verilog, static timing analysis
- ECE-GY 6463 Advanced Hardware Design (recommended)
  - Complex digital logic modeling with RTL
- ECE-GY 9433 Agile System-on-Chip Design ← You are here
  - Bring it all together
  - Complex circuit design using RTL, layout implementation, verification, (simulated) post-silicon validation

Disclaimer

- Third offering of this special topics course
  - More polished than second time
  - But new material, new assignments
- Consider the syllabus more aspirational than concrete
- Dates, assignments subject to change (with advance notice)

Course Overview

Week	Material
1	Syllabus & VLSI Review
2	Agile hardware flows
3-9	SoC Design Process
5-6	**Project proposals**
8	**Midterm**
10-12	System-level considerations
12-13	Advanced design techniques
13	Break
15-16	**Project Presentations**

Core course material

Advanced topics

Course Style

- This is a *bottom-up* class
- Review low-level foundations
- Work up to large-scale, system-level design

Fundamental Elements for Learning

- **Develop foundations** – understand *where* an idea came from
  - Purely for your benefit – will *not* be on the exam
- **Develop intuition** – understand how to compare and synthesize ideas
  - Core of this class – *will* be on the exam
- **Develop experience** – put ideas into practice
  - Core of the project – *best* evaluation of learning

Grading Criteria

- Participation (5%)
  - Based on attendance during lectures and contributions to discussions
- Assignments (18%)
  - Roughly weekly during first half of semester
  - Homeworks: reinforce most important concepts
  - Labs: develop components used towards your course project
- Exam (33%)
- SoC design project (44%)

Attendance Policy

- Points for showing up to lectures / tutorials and participating
- Be on time; stay until the end
- Please stay home or mask if you are sick
- Two excused absences, no questions asked
  - Afterwards, participation grade affected
  - For additional illness or extenuating circumstances, please contact me
- I am generally reasonable, even more so if you approach me beforehand

Attendance Questions

- Attendance points for simple “quiz” questions on Brightspace
- Question 1: answer a question within first 5 min of class
- Question 2: answer a question at end of class
- Questions will be on the projector; submit answer on Brightspace
- Next week (Sep. 9) will be “practice questions” – not graded
  - Start grading the following week (Sep. 16)
- No post-facto excuses accepted
  - “I clicked wrong”, “I was actually here”, “My WiFi wasn’t working”, etc.

Participation Points

- ~25 lectures = ~50 questions = ~2% participation per question
- “2 excused absences” really means **drop lowest 4 questions**
- Additional credit for participation in lectures and tutorials
  - Mainly asking good questions
  - Not recorded in gradebook – do not ask or inquire about points

Course Project

- Majority of this course is a group project
- Groups of 3 students
- Design a specialized, tape-out ready SoC

NDA Materials

- This class will (hopefully) be using confidential materials from TSMC
- You must sign a non-disclosure agreement in order to complete labs and the course project
- If you do not agree to the NDA, you should drop this course before the add / drop deadline
- Any concerns: please contact me

IMPORTANT

Textbooks

> For reference: dense material

- *CMOS VLSI Design: A Circuits and Systems Perspective* (Weste & Harris)
- *VLSI Physical Design: From Graph Partitioning to Timing Closure* (Kahng et al.)
- Both are optional, but highly recommended

better for learning.

Piazza

- Question and Answer board for classes
- All instructors will see questions and can respond
- Can post anonymously (hide your name)
- Can post privately (instructors only) or publicly (+students)
  - Only mark private for personal info. Others may have same question
  - Private questions will be made public if no reason for private
- **Primary interface for communication with instructors**
  - Only send me email for personal issues which cannot be handled by Cas
    - Include “[ECE 9443]” in subject – may get lost otherwise

Assignment Collaboration

- You are individually responsible for understanding material covered in assignments
- It is expected that you at a minimum *attempt* to complete the entire assignment on your own <u>without any outside help</u>
- Once you have given the assignment your best effort, you are welcome and <u>encouraged</u> to discuss your solutions with others in order to verify your understanding and affirm that your work is correct.

Bug Bounty

- We can make mistakes
- You can get extra credit for finding them
  - Homeworks, labs, and course infrastructure (i.e. autograder) only
- See syllabus for more info

Policy on Generative AI

- It is the future
- You should use it
- Be vigilant for hallucinations
- You are encouraged to share experiences
- We will test assignments against generative AIs so that they aren't trivial
  - If you do find an AI that makes a question trivial, let me know
- No limitations on using it for the course project!

"Agile System-on-Chip Design"

See Syllabus for More Info

- Academic Honesty Policy
- Disability accommodations
- Detailed course topics
- Up-to-date assignment due dates

Acknowledgement

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